## 2.7 Abelian Categories

Following [Fre64] we define an Abelian category without reference to additivity, however to make the presentation simpler we assume finite limits and colimits exist.

**Definition 2.7.1**
*We say a category $\mathcal{C}$ is Abelian if it*

*a) is pointed,*
*b) has finite limits and colimits, and*
*c) is binormal*

**Remark 2.7.2 (Abelian is Self-Dual)**
*If $\mathcal{C}$ is Abelian then so is $\mathcal{C}^{op}$. Indeed $\mathcal{C}^{op}$ is again pointed with the same zero object (2.6.54); finite limits in $\mathcal{C}$ become finite colimits in $\mathcal{C}^{op}$ and vice versa (2.6.92), so $\mathcal{C}^{op}$ again has finite limits and colimits; and $\mathcal{C}^{op}$ is binormal, since monic/epic swap under passing to $\mathcal{C}^{op}$ (2.6.33) while $\ker(f)/\mathrm{coker}(f)$ swap (2.6.154), so a monomorphism being a kernel in $\mathcal{C}$ corresponds to an epimorphism being a cokernel in $\mathcal{C}^{op}$ and vice versa.*

**Lemma 2.7.3**
*Let $\mathcal{C}$ be a normal pointed category and $m : I \hookrightarrow Y$ a monomorphism with cokernel $p : Y \to \mathrm{coker}(m)$. Then $f : X \to Y$ factors through $m$ if and only if $p \circ f = 0$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Suppose $f = m \circ g$ for some $g : X \to I$. Then $p \circ f = p \circ m \circ g = 0 \circ g = 0$, since $p \circ m = 0$ by definition of the cokernel (2.6.153).

Conversely, suppose $p \circ f = 0$. As $\mathcal{C}$ is normal and $m$ is monic, $m$ is a normal monomorphism (2.6.163), i.e. $(m, I)$ is the kernel of some morphism (2.6.159). By (2.6.162) it follows that $m = \ker(\mathrm{coker}(m)) = \ker(p)$. So by (2.6.150), $p \circ f = 0$ gives a (unique) morphism $g : X \to I$ such that $m \circ g = f$, that is, $f$ factors through $m$. ∎

**Lemma 2.7.4**
*Let $\mathcal{C}$ be an Abelian category and $f : X \to Y$ a morphism. Then the following are equivalent*

*a) $f$ is epic*
*b) $\mathrm{Im}(f) = Y$ (i.e. $(Y, f, 1_Y)$ is an image for $f$)*
*c) $\mathrm{coker}(f) = 0$*

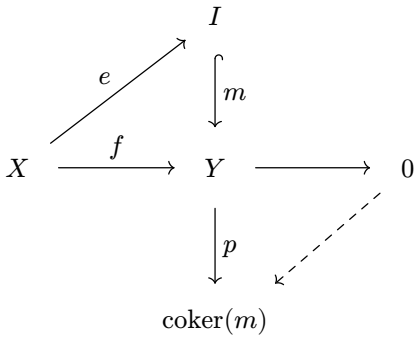
*Dually, the following are equivalent*

*a) $f$ is monic*
*b) $\mathrm{Coim}(f) = X$ (i.e. $(X, 1_X, f)$ is a coimage for $f$)*
*c) $\ker(f) = 0$*

*Proof.* We only prove the first half, with the second following immediately by duality: since $\mathcal{C}$ is Abelian, so is $\mathcal{C}^{op}$ (2.7.2), and by (2.6.169) and (2.6.154) the conditions a)–c) are respectively dual to monic, $\mathrm{Coim}(f) = X$, and $\ker(f) = 0$.

a) $\implies$ c) This is (2.6.151).

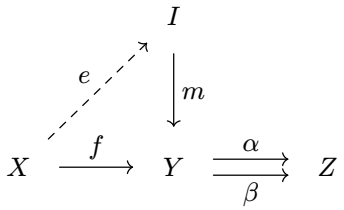
c) $\implies$ b) Suppose $f = m \circ e$ where $m : I \hookrightarrow Y$ is a monomorphism. Consider the cokernel $p : Y \to \mathrm{coker}(m)$




Then $p \circ m = 0 \implies p \circ f = 0$. As $\mathrm{coker}(f) = 0$ there is a morphism $0 \to Z$ such that the bottom right triangle commutes, i.e. $p = 0$. Then $p \circ 1_Y = 0$ and so by (2.7.3) there exists $j : Y \to I$ such that $m \circ j = 1_Y$. As $m$ is monic this factorisation is unique and we conclude that $\mathrm{Im}(f) = Y$.

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b) $\implies$ a) Suppose that $\alpha \circ f = \beta \circ f$ and let $m : \mathrm{eq}(\alpha, \beta) \to Y$ be the equaliser, which is therefore monic (2.6.135).

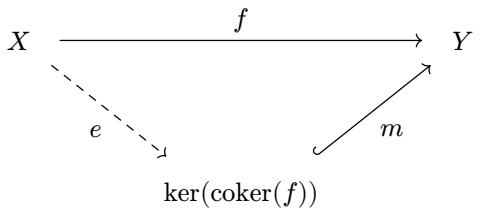



By the universal property of the equaliser there is a (unique) factorisation $f = m \circ e$. Since $(Y, f, 1_Y)$ is the image of $f$ (hypothesis b)) and $(I, e, m)$ is another factorisation of $f$ through a monomorphism, by the universal property of the image (2.6.167) there is a unique morphism $j : Y \to I$ such that $m \circ j = 1_Y$. Therefore $m$ is split-epic so by (2.6.36) it is an isomorphism. Since $(I, m) = \mathrm{eq}(\alpha, \beta)$, by (2.6.136) we conclude $\alpha = \beta$. ∎

### 2.7.1 Images, Coimages and the First Isomorphism Theorem

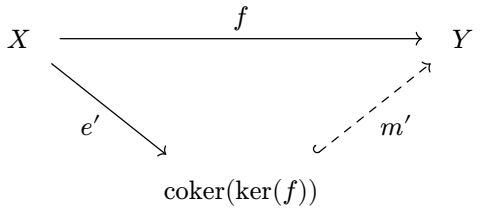
In case of modules we have $\mathrm{coker}(f) = Y/\mathrm{Im}(f)$ and $\ker(\mathrm{coker}(f)) = \mathrm{Im}(f)$ so we might expect this relation to hold in the abstract setting.

**Proposition 2.7.5 ($\mathrm{Im}(f) = \ker(\mathrm{coker}(f))$ and $\mathrm{Coim}(f) = \mathrm{coker}(\ker(f))$)**
*Let $\mathcal{C}$ be a pointed category with kernels and cokernels and $f : X \to Y$ a morphism. Then there is a unique morphism $e : X \to \ker(\mathrm{coker}(f))$ making the following diagram commute*




*When $\mathcal{C}$ is Abelian $e$ is an epimorphism and $\mathrm{Im}(f) = (\ker(\mathrm{coker}(f)), e, m)$.*

*Dually, there is a unique morphism $m' : \mathrm{coker}(\ker(f)) \to Y$ such that the following diagram commutes*

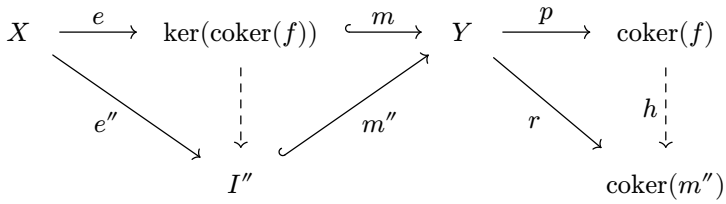



*When $\mathcal{C}$ is Abelian $m'$ is a monomorphism and $\mathrm{Coim}(f) = (\mathrm{coker}(\ker(f)), e', m')$.*

*Proof.* Let $p : Y \to \mathrm{coker}(f)$ be the cokernel of $f$, so by (2.6.153) $p \circ f = 0$. Then by (2.6.150), applied to the kernel $m = \ker(p)$, $e$ exists and is unique.

Dually, let $i : \ker(f) \to X$ be the kernel of $f$, so by (2.6.150) $f \circ i = 0$. Then by (2.6.153), applied to the cokernel $e' = \mathrm{coker}(i)$, $m'$ exists and is unique.

Now suppose additionally that $\mathcal{C}$ is Abelian. We first show that $(\ker(\mathrm{coker}(f)), e, m) = \mathrm{Im}(f)$. By (2.6.135) $m$ is a monomorphism. Consider another factorisation $f = m'' \circ e''$ with $m''$ a monomorphism




Let $r := \mathrm{coker}(m'')$. Then $r \circ f = r \circ m'' \circ e'' = 0$. Therefore, by the universal property of $\mathrm{coker}(f)$ (2.6.153), there is a unique morphism $h : \mathrm{coker}(f) \to \mathrm{coker}(m'')$ such that $h \circ p = r$, and we deduce that $r \circ m = h \circ p \circ m = h \circ 0 = 0$. Therefore by (2.7.3) $m$ factors through $m''$ as required. Uniqueness follows because $m''$ is monic and therefore $(\ker(\mathrm{coker}(f)), e, m)$ is an image for $f$

Next we show $e$ is an epimorphism. By (2.7.4) applied to $e$ (with codomain $I := \ker(\mathrm{coker}(f))$), it suffices to show $\mathrm{Im}(e) = I$, i.e. that $(I, e, 1_I)$ is the image of $e$. Suppose $e = m_1 \circ e_1$ for some monomorphism $m_1 : I_1 \to I$. Then $f =$

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$m \circ e = (m \circ m_1) \circ e_1$ is a factorisation of $f$ through the monomorphism $m \circ m_1$, so by the universal property of the image $(I, e, m)$ of $f$ (2.6.167) (established above) there is a unique morphism $g : I \to I_1$ such that $(m \circ m_1) \circ g = m$; as $m$ is monic this gives $m_1 \circ g = 1_I$. This is precisely the factorisation required for $(I, e, 1_I)$ to be the image of $e$, so $\mathrm{Im}(e) = I$, and therefore $e$ is an epimorphism.

The remaining claims for the dual statement now follow by duality, applying the above to $f^{\mathrm{op}}$ in $\mathcal{C}^{\mathrm{op}}$, which is again Abelian (2.7.2): by (2.6.154), $\ker_{\mathcal{C}^{\mathrm{op}}}(\mathrm{coker}_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}}))$ is identified with $\mathrm{coker}(\ker(f))$, matching (by uniqueness of the first paragraph, dualized) our $m'$ and $e'$, and by (2.6.33) and (2.6.169) the epic/image conclusions in $\mathcal{C}^{\mathrm{op}}$ translate to $m'$ monic and $(\mathrm{coker}(\ker(f)), e', m') = \mathrm{Coim}(f)$ in $\mathcal{C}$. ∎

In the category of modules we have $\mathrm{coker}(\ker(f)) = X/\ker(f)$ and a canonical isomorphism $X/\ker(f) \xrightarrow{\sim} \mathrm{Im}(f) = \ker(\mathrm{coker}(f))$, so we may expect something similar to hold true in the Abelian setting.

**Proposition 2.7.6 (First Isomorphism Theorem)**
*Let $\mathcal{C}$ be a pointed category with kernels and cokernels and $f : X \to Y$ a morphism. Then there is a unique morphism $\theta : \mathrm{coker}(\ker(f)) \to \ker(\mathrm{coker}(f))$ such that the following diagram commutes*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ \downarrow{\scriptstyle e'} & & \uparrow{\scriptstyle m} \\ \mathrm{coker}(\ker(f)) & \xrightarrow{\quad \theta \quad} & \ker(\mathrm{coker}(f)) \end{array}$$

*When $\mathcal{C}$ is an Abelian category then $\theta$ is an isomorphism. Furthermore both $(\mathrm{coker}(\ker(f)), e', m \circ \theta)$ and $(\ker(\mathrm{coker}(f)), \theta \circ e', m)$ are epi-mono factorisations of $f$ and both an image and coimage.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (2.7.5) there is a unique morphism $m' : \mathrm{coker}(\ker(f)) \to Y$ such that $f = m' \circ e'$.

Since $q \circ f = 0$, where $q : Y \to \mathrm{coker}(f)$ is the cokernel of $f$ (2.6.153), we have $q \circ m' \circ e' = q \circ f = 0$; as $e'$ is epic (2.6.135), $q \circ m' = 0$. So by the universal property of $m = \ker(q)$ (2.6.150), there is a unique morphism $\theta : \mathrm{coker}(\ker(f)) \to \ker(\mathrm{coker}(f))$ such that $m \circ \theta = m'$. Combining, $m \circ \theta \circ e' = m' \circ e' = f$, as required.

For uniqueness, suppose $\theta'$ also satisfies $m \circ \theta' \circ e' = f$. Then $m \circ (\theta \circ e') = f = m \circ (\theta' \circ e')$, and since $m$ is monic (2.6.135), $\theta \circ e' = \theta' \circ e'$; as $e'$ is epic (2.6.135), $\theta = \theta'$.

Suppose now $\mathcal{C}$ is Abelian, so that $f = m \circ e$ is the image factorisation and $f = m' \circ e'$ is the coimage factorisation (2.7.5). Since $m \circ \theta \circ e' = f = m \circ e$ and $m$ is monic (2.6.135), we may conclude that $\theta \circ e' = e$.

Since $m \circ \theta = m'$ is monic, $\theta$ is monic, and since $\theta \circ e' = e$ is epic, $\theta$ is epic. As $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so $\theta$, being both monic and epic, is an isomorphism.

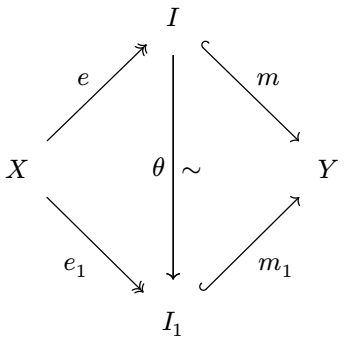
Finally, applying (2.6.172) to the coimage $(\mathrm{coker}(\ker(f)), e', m')$, and dually to the image $(\ker(\mathrm{coker}(f)), e, m)$, via the isomorphism $\theta$ shows each factorisation is both an image and a coimage of $f$. ∎

**Proposition 2.7.7 (Epi-Mono Factorisation in an Abelian Category)**
*Let $\mathcal{C}$ be an Abelian category and $f : X \to Y$ a morphism. Then $f$ factors as $f = m \circ e$ for some epimorphism $e$ and monomorphism $m$*

$$\begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & {\scriptstyle e}\searrow & & \nearrow{\scriptstyle m} & \\ & & I & & \end{array}$$

*and this factorisation is unique up to isomorphism: if $f = m_1 \circ e_1$ is another such factorisation, with $e_1 : X \to I_1$ epic and $m_1$ monic, then there is a unique morphism $\theta : I \to I_1$ such that the following diagram commutes, and $\theta$ is (moreover) an isomorphism*

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*Moreover, every epimorphism and every monomorphism of $\mathcal{C}$ is regular, so any such factorisation $f = m_1 \circ e_1$ is simultaneously a regular epi-mono and an epi-regular-mono factorisation of $f$, and hence $(I_1, e_1, m_1)$ is both an image and a coimage of $f$ (2.6.174).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**Regularity.** Since $\mathcal{C}$ is Abelian it is binormal (2.7.1), so every monomorphism is normal and every epimorphism is conormal (2.6.163), hence every monomorphism and every epimorphism of $\mathcal{C}$ is regular (2.6.160). In particular, for any epi-mono factorisation $f = m_1 \circ e_1$, $e_1$ is a regular epimorphism and $m_1$ is a regular monomorphism, so $f = m_1 \circ e_1$ is simultaneously a regular epi-mono and an epi-regular-mono factorisation of $f$, and by (2.6.174) (both halves), $(I_1, e_1, m_1)$ is both an image and a coimage of $f$.

**Existence.** By (2.7.5), taking $I := \ker(\operatorname{coker}(f))$ with structure maps $e, m$, we obtain a factorisation $f = m \circ e$ with $m$ monic and, since $\mathcal{C}$ is Abelian, $e$ epic.

**Uniqueness up to isomorphism.** Let $f = m_1 \circ e_1$ be any other such factorisation, $e_1 : X \to I_1$. By the Regularity paragraph, $e$ and $e_1$ are regular epimorphisms, so $f = m \circ e$ and $f = m_1 \circ e_1$ are regular epi-mono factorisations of $f$ (2.6.173), and (2.6.178) gives a unique isomorphism $\theta : I \to I_1$ with $\theta \circ e = e_1$ and $m_1 \circ \theta = m$.

*Alternatively*, uniqueness may be shown directly from the universal property of the image (2.6.167), without invoking regular epi-mono factorisations at all.

**Uniqueness up to isomorphism.** Let $f = m_1 \circ e_1$ be any other epi-mono factorisation, $e_1 : X \to I_1$. Since $(I, e, m) = \operatorname{Im}(f)$ (2.7.5) and $m_1$ is monic, the universal property of the image (2.6.167) gives a unique morphism $\theta : I \to I_1$ such that $m_1 \circ \theta = m$, and (as noted there) $\theta \circ e = e_1$.

We show $\theta$ is an isomorphism. If $\theta k_1 = \theta k_2$ then $m_1 \theta k_1 = m_1 \theta k_2$, i.e. $m k_1 = m k_2$, and since $m$ is monic, $k_1 = k_2$; so $\theta$ is monic. Since $\theta \circ e = e_1$ is epic, $\theta$ is epic. As $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so $\theta$, being both monic and epic, is an isomorphism. ∎

## 2.7.2 Exact Sequences

**Lemma 2.7.8 (Cancellation for Kernels and Cokernels)**
*Let $\mathcal{C}$ be a pointed category with kernels and cokernels.*

*a) If $h = \psi \circ p$ with $\psi$ monic, then $(K, i)$ is a kernel of $p$ if and only if it is a kernel of $h$.*
*b) If $f = m \circ e$ with $e$ epic, then $(Q, q)$ is a cokernel of $m$ if and only if it is a cokernel of $f$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

a) For any $k$ into the common domain of $h, p$, $h \circ k = \psi \circ p \circ k$; since $\psi$ is monic, $\psi \circ p \circ k = 0 \iff p \circ k = 0$, so $h \circ k = 0 \iff p \circ k = 0$. In particular $p \circ i = 0 \iff h \circ i = 0$, and for any $i'$, $p \circ i' = 0 \iff h \circ i' = 0$; so the defining conditions of (2.6.150) hold for $(K, i)$ and $p$ if and only if they hold for $(K, i)$ and $h$, that is, $(K, i)$ is a kernel of $p$ if and only if it is a kernel of $h$.

b) Dually, for any $q'$ out of the common codomain of $f, m$, $q' \circ f = q' \circ m \circ e$; since $e$ is epic, $q' \circ m \circ e = 0 \iff q' \circ m = 0$, so $q' \circ f = 0 \iff q' \circ m = 0$; in particular $q \circ m = 0 \iff q \circ f = 0$, and for any $q'$, $q' \circ m = 0 \iff q' \circ f = 0$; so the defining conditions of (2.6.153) hold for $(Q, q)$ and $m$ if and only if they hold for $(Q, q)$ and $f$, that is, $(Q, q)$ is a cokernel of $m$ if and only if it is a cokernel of $f$. ∎

**Proposition 2.7.9 (Criteria for Exactness)**
*Let $\mathcal{C}$ be an Abelian category, $X \xrightarrow{f} Y \xrightarrow{g} Z$ a pair of composable morphisms, $f = m \circ e$ an epi-mono factorisation of $f$ and $g = m' \circ e'$ an epi-mono factorisation of $g$. Then the following are equivalent:*

*a) $m = \operatorname{Im}(f) = \ker(g)$ (i.e. exactness in the sense of (2.6.200));*

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*b)* $g \circ f = 0$*, and the canonical morphism* $\varphi : \mathrm{Im}(f) \to \ker(g)$ *is epic;*
*c)* $\mathrm{coker}(f) = \mathrm{Coim}(g) = e'$*.*
*d)* $g \circ f = 0$*, and the canonical morphism* $\varphi : \mathrm{coker}(f) \to \mathrm{Coim}(g)$ *is monic.*

*This does not depend on the choice of factorisation of either* $f$ *or* $g$*.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Write $k : \ker(g) \to Y$ for a kernel of $g$.

**Independence of factorisation of $f$.** By (2.6.200) (independence of representatives), applied with $\mathcal{C}$ Abelian – so every epi-mono factorisation of $f$ is (in particular) an image of $f$ (2.7.7) – exactness of the pair at $Y$, in particular the condition $m = \ker(g)$, does not depend on the choice of epi-mono factorisation of $f$: if $f = m_1 \circ e_1$ is another, then $m = \ker(g)$ if and only if $m_1 = \ker(g)$.

**Independence of factorisation of $g$.** If $g = m_1' \circ e_1'$ is another epi-mono factorisation of $g$, (2.7.7) gives an isomorphism $\phi : I' \to I_1'$ with $\phi \circ e' = e_1'$, so by (2.6.138) (applied to $\phi$ and to $\phi^{-1}$), $\mathrm{coker}(f) = e'$ if and only if $\mathrm{coker}(f) = e_1'$.

**Preliminary.** By (2.7.7), $(I', e', m')$ is both an image and a coimage of $g$; in particular $\mathrm{Coim}(g) = e'$.

a) $\Longrightarrow$ b) Taking $\varphi := 1_I$ gives $k \circ \varphi = k = m$; as $g \circ k = 0$ (2.6.150), $g \circ f = g \circ m \circ e = g \circ k \circ e = 0$. So (by uniqueness) $\varphi = 1_I$ is the morphism of b), and is epic, being an isomorphism.

b) $\Longrightarrow$ a) Since $m = k \circ \varphi$ and $m$ is monic, $\varphi$ is monic: if $\varphi \circ a = \varphi \circ b$ then $m \circ a = k \circ \varphi \circ a = k \circ \varphi \circ b = m \circ b$, so $a = b$. As $\varphi$ is also epic (b)), and $\mathcal{C}$ is Abelian hence balanced (2.6.164), $\varphi$ is an isomorphism; so $m = k \circ \varphi$ exhibits $m = \ker(g)$, that is, a).

a) $\Longrightarrow$ c) By (2.7.8) b) (as $e$ is epic), $\mathrm{coker}(f) = \mathrm{coker}(m)$; since $m = \ker(g)$ (a)), $\mathrm{coker}(m) = \mathrm{coker}(\ker(g)) = \mathrm{Coim}(g)$ (2.7.5), which by the Preliminary equals $e'$. So $\mathrm{coker}(f) = e'$, that is, c).

c) $\Longrightarrow$ a) By hypothesis $\mathrm{coker}(f) = e'$; since $g = m' \circ e'$, (2.7.8) a) (with $m'$ in the role of the monomorphism and $e' = \mathrm{coker}(f)$ in the role of $p$) gives $\ker(g) = \ker(\mathrm{coker}(f))$. By (2.7.5), $\mathrm{Im}(f) = \ker(\mathrm{coker}(f))$, i.e. (as $\mathrm{Im}(f) = (I, e, m)$) $m = \ker(\mathrm{coker}(f)) = \ker(g)$, that is, a).

c) $\Longleftrightarrow$ d) By duality: apply a) $\Longleftrightarrow$ b), established above for an arbitrary composable pair in an Abelian category, to $g^{\mathrm{op}} : Z \to Y$ and $f^{\mathrm{op}} : Y \to X$ in $\mathcal{C}^{\mathrm{op}}$ (Abelian, by (2.7.2)), using the epi-mono factorisation $g^{\mathrm{op}} = (e')^{\mathrm{op}} \circ (m')^{\mathrm{op}}$ of $g^{\mathrm{op}}$ (dual to $g = m' \circ e'$). This gives: $(e')^{\mathrm{op}} = \ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ if and only if $f^{\mathrm{op}} \circ g^{\mathrm{op}} = 0$ and the canonical morphism $\mathrm{Im}_{\mathcal{C}^{\mathrm{op}}}(g^{\mathrm{op}}) \to \ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ is epic in $\mathcal{C}^{\mathrm{op}}$.

Translating via (2.6.154) ($\ker_{\mathcal{C}^{\mathrm{op}}}(f^{\mathrm{op}})$ corresponds to $\mathrm{coker}(f)$), (2.6.169) ($\mathrm{Im}_{\mathcal{C}^{\mathrm{op}}}(g^{\mathrm{op}})$ corresponds to $\mathrm{Coim}(g)$), and (2.6.33) (epic in $\mathcal{C}^{\mathrm{op}}$ corresponds to monic in $\mathcal{C}$), while $f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$ vanishes if and only if $g \circ f = 0$: this reads exactly as $e' = \mathrm{coker}(f)$ if and only if $g \circ f = 0$ and the canonical morphism $\mathrm{coker}(f) \to \mathrm{Coim}(g)$ is monic, that is, c) $\Longleftrightarrow$ d).

**Independence of choice.** a) does not depend on the choice of epi-mono factorisation of $f$ (shown above), nor of $g$ (as a) only references the fixed kernel $k$, not the chosen factorisation of $g$); so, by the equivalences just established, neither do b), c), d), each being equivalent to a). ∎

**Remark 2.7.10**
*Conditions b)–d) give further equivalent characterisations of this same notion of exactness at* $Y$*, in particular the self-dual form* $\mathrm{coker}(f) = \mathrm{Coim}(g)$*.*

## 2.7.3 Biproducts

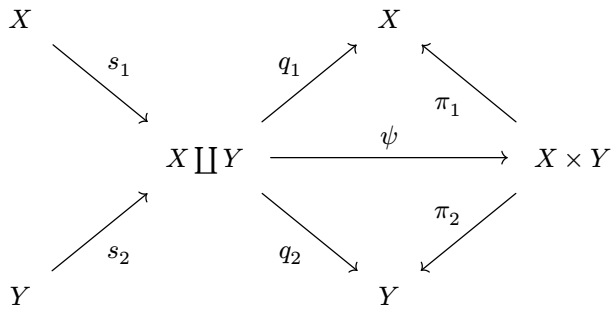
**Lemma 2.7.11**
*Let* $\mathcal{C}$ *be an Abelian category,* $X, Y \in \mathcal{C}$*, and* $\psi : X \coprod Y \to X \times Y$ *the canonical comparison morphism (2.6.104), where* $\pi_1, \pi_2$ *are the projections of* $X \times Y$ *and* $s_1, s_2$ *are the injections of* $X \coprod Y$*. Then* $\psi$ *is monic.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

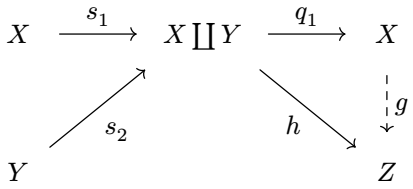
Write $X_1 := X$, $X_2 := Y$.

**Maps out of the coproduct.** By (2.6.93) let $q_1 := [1, 0] : X \coprod Y \to X$ and $q_2 := [0, 1] : X \coprod Y \to Y$, i.e. the morphisms induced by $q_k \circ s_i = \delta_{ki}$. Since $\pi_i \circ \psi \circ s_j = \delta_{ij} = q_i \circ s_j$ for all $j$, by the uniqueness clause of (2.6.93), $\pi_i \circ \psi = q_i$, that is, the following diagram commutes (which also displays the canonical injections $s_1, s_2$)

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**$\operatorname{coker}(s_1) = q_2$ and $\operatorname{coker}(s_2) = q_1$.** Indeed $q_1 s_2 = \delta_{12} = 0$ by construction. By (2.6.153), given $h : X \coprod Y \to Z$ with $h s_2 = 0$, we must produce a unique $g : X \to Z$ making the following diagram commute

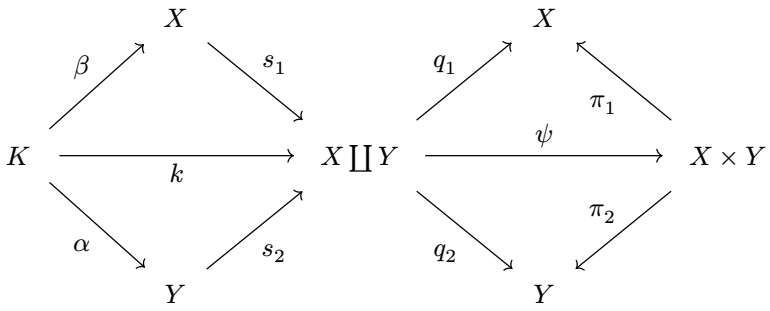



By the uniqueness clause of (2.6.93), $h = [h s_1, h s_2]$ and, for any $g : X \to Z$, $g \circ [1, 0] = [g, 0]$ (both sides restrict to the same maps along $s_1, s_2$). So since $h s_2 = 0$, $h = [h s_1, 0] = (h s_1) \circ [1, 0] = (h s_1) \circ q_1$. Setting $g := h s_1$ then ensures the diagram commutes. Since $q_1 s_1 = \delta_{11} = 1_X$, $q_1$ is split-epic, hence epic (2.6.35), so $g$ is the unique such morphism. Therefore $\operatorname{coker}(s_2) = q_1$ as required. By the symmetric argument, $\operatorname{coker}(s_1) = q_2$.

**$s_1 = \ker(q_2)$ and $s_2 = \ker(q_1)$.** Since $q_1 s_1 = \delta_{11} = 1_X$, $s_1$ is split-monic, hence monic (2.6.35); similarly $s_2$ is monic (via $q_2 s_2 = \delta_{22} = 1_Y$). As $\mathcal{C}$ is Abelian it is binormal (2.7.1), so the monomorphism $s_2$ is normal (2.6.163), i.e. the kernel of some morphism (2.6.159), whence by (2.6.162), $s_2 = \ker(\operatorname{coker}(s_2)) = \ker(q_1)$. Similarly $s_1 = \ker(\operatorname{coker}(s_1)) = \ker(q_2)$.

**$\ker(\psi) = 0$.** Since $\pi_1, \pi_2$ jointly determine morphisms into $X \times Y$ (2.6.81), $\psi \circ k = 0$ iff $q_1 k = 0$ and $q_2 k = 0$, that is, $\ker(\psi)$ is the intersection, as subobjects of $X \coprod Y$, of $\ker(q_1) = s_2$ and $\ker(q_2) = s_1$. Informally, $\ker(q_1)$ is the "$Y$-summand" of $X \coprod Y$ (as $q_1$ is the identity on $X$ and kills $Y$) and $\ker(q_2)$ is the "$X$-summand", so this intersection should vanish just as the coordinate subspaces $X \times \{0\}$ and $\{0\} \times Y$ meet only at the origin inside $X \oplus Y$; we verify this directly.

Suppose $k : K \to X \coprod Y$ satisfies $\psi \circ k = 0$. Then $q_1 k = \pi_1 \psi k = 0$ and $q_2 k = \pi_2 \psi k = 0$. Since $s_2 = \ker(q_1)$ there is $\alpha : K \to Y$ with $s_2 \alpha = k$, and since $s_1 = \ker(q_2)$ there is $\beta : K \to X$ with $s_1 \beta = k$, so $s_2 \alpha = s_1 \beta$, that is, the following diagram commutes




Composing with $q_1$: $q_1 s_2 \alpha = q_1 s_1 \beta$ gives $0 = \beta$. Therefore $k = s_1 \beta = 0$.

By (2.6.166), $\ker(\psi) = 0$, so by (2.7.4), $\psi$ is monic. ∎

**Proposition 2.7.12 (Finite Biproducts Exist)**
*Let $\mathcal{C}$ be an Abelian category. Then $\mathcal{C}$ has all finite biproducts (2.6.103).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (2.6.105) it suffices to show that binary biproducts exist. Fix $X, Y \in \mathcal{C}$; by (2.6.104) there is a unique comparison morphism $\psi : X \coprod Y \to X \times Y$, with $\pi_1, \pi_2$ the projections of $X \times Y$ and $s_1, s_2$ the injections of $X \coprod Y$, and it suffices to show $\psi$ is an isomorphism.

By (2.7.11), $\psi$ is monic. Since $\mathcal{C}^{\mathrm{op}}$ is again Abelian (2.7.2), in which $X \times Y$ (with $\pi_1, \pi_2$) and $X \coprod Y$ (with $s_1, s_2$) play the roles of the coproduct and product respectively (2.6.92), the canonical comparison morphism for $X, Y$ in $\mathcal{C}^{\mathrm{op}}$ is

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exactly $\psi^{\mathrm{op}}$, since both are characterised by the same defining equations $\pi_i \circ (-) \circ s_j = \delta_{ij}$. So applying (2.7.11) to $\mathcal{C}^{\mathrm{op}}$, $\psi^{\mathrm{op}}$ is monic in $\mathcal{C}^{\mathrm{op}}$, i.e. $\psi$ is epic in $\mathcal{C}$ (2.6.33).

As $\mathcal{C}$ is binormal it is balanced (2.6.164), so $\psi$, being both monic and epic, is an isomorphism. By (2.6.104), a biproduct $X \oplus Y$ therefore exists. ∎

**Proposition 2.7.13 ($\mathrm{Mor}(X,Y)$ is an Abelian Group)**
*Let $\mathcal{C}$ be an Abelian category and $X, Y \in \mathcal{C}$. Then every $f \in \mathrm{Mor}(X,Y)$ has an additive inverse; combined with (2.6.121) (applicable since $\mathcal{C}$ has all finite biproducts (2.7.12)), $(\mathrm{Mor}(X,Y), +)$ is an abelian group.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.* Following [Fre64] Theorem 2.39.

The underlying idea: in a category of modules, the inverse of the unitriangular matrix $\left(\begin{smallmatrix} 1 & 0 \\ f & 1 \end{smallmatrix}\right)$ is $\left(\begin{smallmatrix} 1 & 0 \\ -f & 1 \end{smallmatrix}\right)$, so the additive inverse of $f$ can be read off from the inverse of this matrix; we therefore try to show that the same matrix $M$ is an isomorphism in any Abelian category.

Fix $f : X \to Y$ and let $M := \left(\begin{smallmatrix} 1_X & 0 \\ f & 1_Y \end{smallmatrix}\right) : X \oplus Y \to X \oplus Y$ (2.6.108). By (2.6.122), matrices now multiply in the usual way, entry-by-entry, dot-product style.

*$M$ is monic.* For $g = \left(\begin{smallmatrix} g_1 \\ g_2 \end{smallmatrix}\right) : Z \to X \oplus Y$

$$M \circ g = \begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} \circ \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \begin{pmatrix} g_1 \\ f \circ g_1 + g_2 \end{pmatrix}$$

so $M \circ g = 0$ gives $g_1 = 0$ and $f \circ g_1 + g_2 = 0$; substituting, $f \circ 0 + g_2 = 0$, i.e. $g_2 = 0$, so $g = 0$. By (2.6.166), $\ker(M) = 0$, so by (2.7.4), $M$ is monic.

*$M$ is epic.* Dually, for $h = \left(\begin{smallmatrix} h_1 & h_2 \end{smallmatrix}\right) : X \oplus Y \to Z$

$$h \circ M = \begin{pmatrix} h_1 & h_2 \end{pmatrix} \circ \begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} = \begin{pmatrix} h_1 + h_2 \circ f & h_2 \end{pmatrix}$$

so $h \circ M = 0$ gives $h_2 = 0$ and $h_1 + h_2 \circ f = 0$, i.e. $h_1 = 0$, so $h = 0$. Dually, $\mathrm{coker}(M) = 0$, so by (2.7.4), $M$ is epic.

*$M$ is invertible.* Since $\mathcal{C}$ is Abelian it is binormal (2.7.1), hence balanced (2.6.164), so $M$, being monic and epic, is an isomorphism.

*Extracting the inverse.* Write $M^{-1} = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$. Then

$$\begin{pmatrix} 1_X & 0 \\ f & 1_Y \end{pmatrix} \circ \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p & q \\ f \circ p + r & f \circ q + s \end{pmatrix} = \begin{pmatrix} 1_X & 0 \\ 0 & 1_Y \end{pmatrix}$$

gives $p = 1_X$, $q = 0$, $f \circ p + r = 0$ and $f \circ q + s = 1_Y$. Substituting $p = 1_X$ into the third equation gives $f + r = 0$: so $f$ has additive inverse $-f := r$. ∎

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